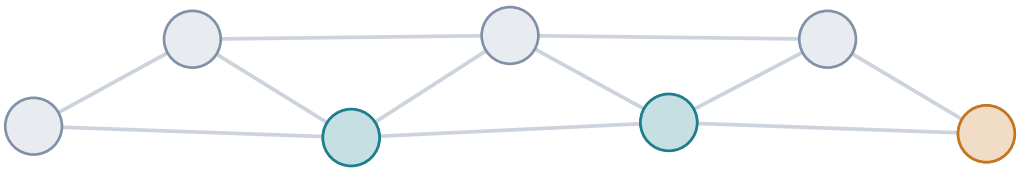
[COLLEGE / INSTITUTE NAME]

[Department Name]

LABORATORY MANUAL & RECORD

COMPUTER NETWORKS LABORATORY

Network Simulation with NS-2 • Data-Link Layer Programs in Python



Name	[Student Name]
USN / Roll No.	[USN / Roll No.]
Semester	[Semester]
Section / Batch	[Section / Batch]
Academic Year	2026–27
Faculty In-charge	[Faculty In-charge]

Experiments 1–6 • NS-2 • AWK • XGraph • Python

[COLLEGE / INSTITUTE NAME]
[Department Name]

C E R T I F I C A T E

This is to certify that **[Student Name]**, bearing USN / Roll No. **[USN / Roll No.]**, of [Semester] semester ([Section / Batch]), has satisfactorily completed all the experiments prescribed for the **Computer Networks Laboratory** in the academic year 2026–27, as recorded in this manual.

[Faculty In-charge]
Faculty In-charge

Head of the Department
[Department Name]

Date: _____

Place: _____

Index

Sl. No.	Date	Name of the experiment	Page	Marks	Signature
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2	21/08/2026	Four-node network with TCP and UDP traffic: packet counts and throughput	7		
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Appendix A (viva voce question bank) is on page 30; Appendix B (troubleshooting guide) is on page 32; Appendix C (files and quick command reference) is on page 33.

INTRODUCTION

NS-2 and the Experimental Workflow

This manual records six experiments of the **Computer Networks Laboratory**. Experiments 1–4 are network simulations written in Tcl for the **NS-2** simulator and analysed with **AWK** scripts and **XGraph** plots; Experiments 5 and 6 are data-link layer programs (bit stuffing and CRC) written in **Python**. Every experiment follows the same structure: aim, theory, topology or algorithm, program, execution, output and result; the viva voce questions of all experiments are collected in Appendix A.

Software environment

Tool	Version	Purpose
Operating system	Ubuntu (Linux)	Lab workstation; all commands are run from a terminal
NS-2	2.35	Discrete-event network simulator, started with `ns script.tcl`
NAM	1.15	Network Animator; replays the `.nam` file written by NS-2
XGraph	4.38	Plots two-column (`x y`) data files
AWK	mawk / gawk	Post-processes the trace file (counts, throughput, congestion window)
Python	3.x	Programs of Experiments 5 and 6

Simulation workflow

An NS-2 simulation is described by a Tcl script. Running it produces two files: a *trace file* (`.tr`) that records every packet event, and a *NAM file* (`.nam`) used to animate the simulation. An AWK script reduces the trace file to the numbers or the data file that is finally plotted with XGraph.

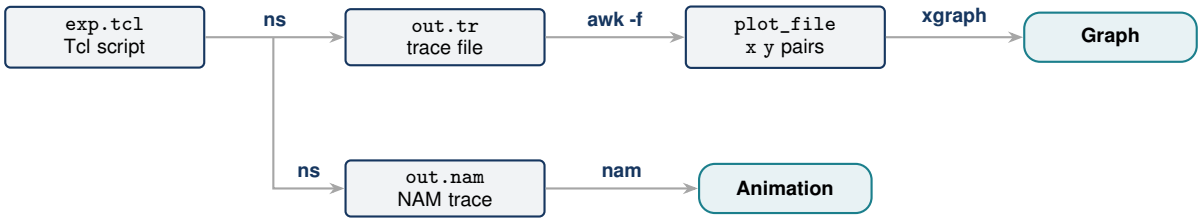


Figure 1. From Tcl script to results: the files and tools used in Experiments 1–4.

Reading the NS-2 trace file

Each line of the trace file describes one packet event. The first lines of the trace produced by Experiment 1 are shown below, followed by the meaning of the fields. AWK refers to the fields as `$1`, `$2`, ... in this order.

out.tr (Experiment 1, first lines)

```
+ 0.5 0 1 cbr 500 ------- 0 0.0 2.0 0 0
- 0.5 0 1 cbr 500 ------- 0 0.0 2.0 0 0
+ 0.505 0 1 cbr 500 ------- 0 0.0 2.0 1 1
- 0.505 0 1 cbr 500 ------- 0 0.0 2.0 1 1
```

Field	Meaning
`$1`	event: `+` `-` `r` `d` (`h` on a LAN)
`$2`	time of the event (s)
`$3`	from node
`$4`	to node
`$5`	packet type (`cbr`, `tcp`, `ack`)
`$6`	packet size (bytes)

Field	Meaning
`$7`	flags
`$8`	flow id (`class_`)
`$9`	source address
`$10`	destination address
`$11`	sequence number
`$12`	unique packet id

The event symbols are: `+` packet *enqueued*, `-` packet *dequeued* (transmission starts), `r` packet *received* at the far end of the link, `d` packet *dropped*, and `h` a hop event on LAN links.

> **Units used in the scripts.** Bandwidth `1Mb` means 1 Mb/s, `512Kb` means 512 kb/s, delays are in milliseconds (`10ms`), packet sizes are in bytes and all times are in seconds. Throughput computed by the AWK scripts is expressed in kb/s.

Experiments at a glance

No.	Topic	Tools	Key result
1	Point-to-point network: packets dropped against bandwidth and queue size	NS-2, AWK, XGraph	502 drops at 0.25 Mb, none at 1 Mb
2	TCP and UDP traffic: packet counts and throughput	NS-2, AWK, XGraph	TCP 846 to 3582 records (1 to 2 Mb), UDP almost constant
3	Ethernet LAN: data rate and error rate	NS-2, AWK, XGraph	535 to 1369 kb/s for 0.5 to 3 Mb; errors lower the throughput
4	Congestion window of TCP Vegas and Reno	NS-2, AWK, XGraph	Vegas steady at about 22 packets, Reno grows to 30.7
5	Bit stuffing	Python	`1011111101` → `10111110101`
6	CRC error detection	Python	`1001` → codeword `1001110`

EXPERIMENT NO. 1 Date: 14/08/2026

Point-to-Point Network: Queue Size, Bandwidth and Packet Drops

Aim

> Simulate a three-node point-to-point network with duplex links between them. Set the queue size, vary the bandwidth and find the number of packets dropped.

Theory

Duplex links and the DropTail queue

In NS-2 a *duplex link* is a pair of simplex links, one for each direction, each described by a bandwidth, a propagation delay and a queue discipline. The command `$ns duplex-link $n0 $n1 1Mb 10ms DropTail` joins nodes 0 and 1 with a 1 Mb/s link of 10 ms delay. *DropTail* is a first-in-first-out queue with a finite buffer: when the buffer already holds `limit_` packets (50 by default, set explicitly with `queue-limit`), the newly arriving packet is discarded.

Traffic source

A `UDP` agent carries the traffic produced by a `CBR` (constant bit rate) application, and the `Null` agent at the destination simply absorbs the packets. With `packetSize_` = 500 bytes and `interval_` = 0.005 s the source offers

$$R = \frac{500 \times 8 \text{ bits}}{0.005 \text{ s}} = 800 \text{ kb/s}$$

from 0.5 s to 4.5 s, i.e. 800 packets in total.

Why packets are dropped

If the link bandwidth is larger than the offered load (1 Mb/s > 800 kb/s) each packet is transmitted as soon as it arrives and the queue stays empty. If the bandwidth is smaller, the queue grows at the rate $R - C$ (C = link capacity) until it is full, and every further arrival is dropped. The number of drops therefore decreases when the bandwidth or the queue size is increased. A drop is recorded in the trace file as a line beginning with `d`, and the AWK script simply counts these lines.

Network topology

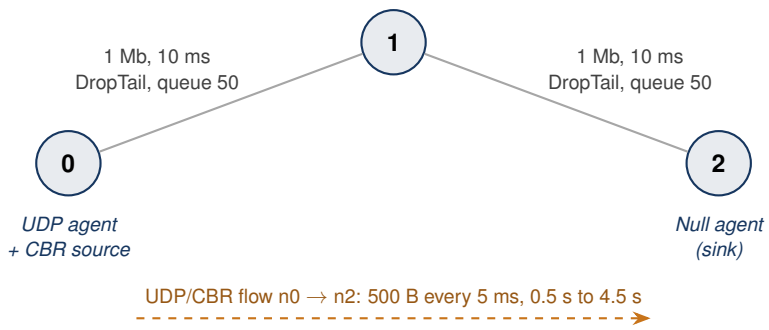


Figure 2. Three-node topology of Experiment 1 (node orientation as set with `duplex-link-op`).

Procedure

1. Create the simulator object and open the NAM file (`out.nam`) and the trace file (`out.tr`).
2. Write the `finish` procedure: flush the trace, close both files and start NAM.
3. Create three nodes `n0`, `n1` and `n2`.
4. Join them with duplex links (1 Mb, 10 ms, DropTail), set the orientation for NAM and set the queue limit of each link to 50 packets.
5. Attach a UDP agent to `n0` and a Null agent to `n2`, and connect them.
6. Attach a CBR application (500 bytes every 0.005 s) to the UDP agent; start it at 0.5 s and stop it at 4.5 s.
7. Schedule `finish` at 5.0 s, run the simulator with `ns exp1.tcl`, and count the `d` events with `awk -f exp1.awk out.tr`.
8. Repeat after changing the bandwidth of the links and, separately, the queue limit; tabulate the drops and plot them with XGraph.

Program

`exp1.tcl` — NS-2 simulation script

```
#gedit exp1.tcl
set ns [new Simulator]

#open a new file for NAMTRACE
set nf [open out.nam w]
$ns namtrace-all $nf

#open a new file to log TRACE
set tf [open out.tr w]
$ns trace-all $tf

#body of finish procedure
proc finish {} {
    global ns nf tf
    $ns flush-trace
    close $nf
    close $tf
    exec nam out.nam &
    exit 0
}

#Create nodes
set n0 [$ns node]
set n1 [$ns node]
set n2 [$ns node]

#Create links between nodes
$ns duplex-link $n0 $n1 1Mb 10ms DropTail
$ns duplex-link $n1 $n2 1Mb 10ms DropTail

#set orientation
$ns duplex-link-op $n0 $n1 orient right-up
$ns duplex-link-op $n1 $n2 orient right-down

#set queue limit-default is 50 packets
$ns queue-limit $n0 $n1 50
$ns queue-limit $n1 $n2 50

#Create transport agent
#Sender
set udp0 [new Agent/UDP]
```

```
$ns attach-agent $n0 $udp0

#Receiver
set null0 [new Agent/Null]
$ns attach-agent $n2 $null0
$ns connect $udp0 $null0

#Create Application to generate traffic
set cbr0 [new Application/Traffic/CBR]
$cbr0 set packetSize_ 500
$cbr0 set interval_ 0.005
$cbr0 attach-agent $udp0

#Start and stop generating traffic
$ns at 0.5 "$cbr0 start"
$ns at 4.5 "$cbr0 stop"

$ns at 5.0 "finish"

$ns run
```

exp1.awk — counts the dropped packets

```
#gedit exp1.awk
BEGIN { count = 0; }
{
    if($1 == "d")
    {
        count++;
        printf("%s\t %s\n", $5, $11);
    }
}
END { printf(" The number of packets dropped =%d \n", count ); }
```

Execution and output

Terminal

```
$ ns exp1.tcl
$ awk -f exp1.awk out.tr
 The number of packets dropped =0
```

With the default bandwidth of 1 Mb the offered load (800 kb/s) fits in the link, so no packet is dropped. If the bandwidth is reduced (here to 0.5 Mb) the script prints the sequence number of every dropped packet and finally the total:

Terminal

```
$ awk -f exp1.awk out.tr        # bandwidth changed to 0.5Mb
cbr      131
cbr      134
cbr      136
...
 The number of packets dropped =252
```

The corresponding drop record in the trace file begins with the event symbol `d`:

out.tr (bandwidth 0.5 Mb)

```
d 1.155 0 1 cbr 500 ------- 0 0.0 2.0 131 131
```

Observations

The simulation was repeated for four bandwidths (queue limit 50) and for four queue limits (bandwidth 0.75 Mb). The counts were stored in the two data files `plot_graph_1` and `plot_graph_2` and plotted with XGraph.

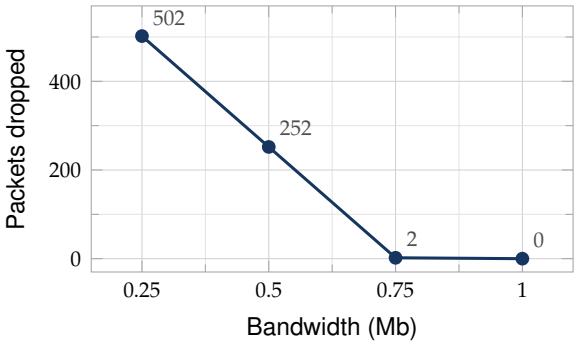
Bandwidth (Mb)	Packets dropped
0.25	502
0.5	252
0.75	2
1	0

Table 1: queue limit = 50 packets (`plot_graph_1`)

Queue limit	Packets dropped
40	12
30	22
20	32
10	42

Table 2: bandwidth = 0.75 Mb (`plot_graph_2`)

Graph 1: bandwidth vs. drop count



Graph 2: queue limit vs. drop count

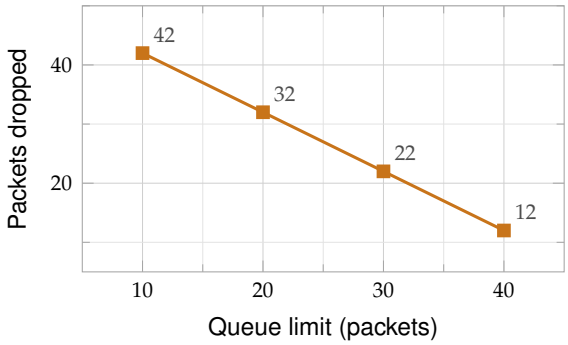


Figure 3. Packets dropped as a function of the link bandwidth and of the queue limit.

- At 1 Mb the link can carry the 800 kb/s offered load, so **no packet is dropped**.
- Reducing the bandwidth to 0.75, 0.5 and 0.25 Mb makes the service rate smaller than the arrival rate, and the drops rise to **2**, **252** and **502** packets.
- At 0.75 Mb a larger queue absorbs more of the excess: the drops fall from **42** (queue of 10 packets) to **12** (queue of 40 packets). A larger buffer reduces loss but increases the queuing delay.

Result. A three-node point-to-point network with duplex links was simulated in NS-2 and the number of dropped packets was determined from the trace file. The number of drops decreases as the link bandwidth or the queue size increases: from 502 packets at 0.25 Mb to none at 1 Mb, and from 42 packets (queue of 10) to 12 packets (queue of 40) at 0.75 Mb.

Viva voce questions for this experiment: see Appendix A, page 30.

EXPERIMENT NO. 2 Date: 21/08/2026

Four-Node Network with TCP and UDP Traffic: Packet Counts and Throughput

Aim

> Simulate a four-node point-to-point network and connect the links as follows: n0–n2, n1–n2 and n2–n3. Apply a TCP agent between n0–n3 and a UDP agent between n1–n3. Apply relevant applications over the TCP and UDP agents, change the parameters, determine the number of packets sent by TCP and UDP and analyse the throughput.

Theory

TCP and UDP in NS-2

UDP is connectionless and unreliable: it has no acknowledgements, no retransmission and no congestion control, so its sending rate is fixed by the application. Here the `CBR` application sends 500 bytes every 5 ms (800 kb/s). **TCP** is connection-oriented and reliable: the receiver acknowledges the data and the sender controls its *window* (slow start and congestion avoidance) so that it adapts to the capacity that is available. In NS-2 the sender is `Agent/TCP`, the receiver is `Agent/TCPSink` (it returns the acknowledgements) and `Application/FTP` supplies an unlimited stream of data to the TCP agent.

Shared bottleneck

Both flows meet at node 2 and share the link n2–n3. Because the UDP source never slows down, TCP obtains only the capacity that is left over, which is why the TCP traffic grows strongly when the link bandwidth is increased while the UDP traffic does not.

Counting packets and computing throughput

`exp2.awk` counts the trace records whose packet type (`$5`) is `cbr` (UDP) or `tcp`. A packet produces several records (enqueue, dequeue and receive on every link it crosses), so the counts are event counts that are used to compare the two protocols and the different bandwidths. `exp2_throughput.awk` adds up the bytes of all received packets (`r` events) and divides by the elapsed time:

$$\text{Throughput} = \frac{\sum \text{bytes of received packets}}{t_{\text{last receive}} - t_{\text{first enqueue}}} \times \frac{8}{1000} \ \text{kb/s}$$

Since a packet is counted at every link it crosses, this is the aggregate throughput of the whole network.

Network topology

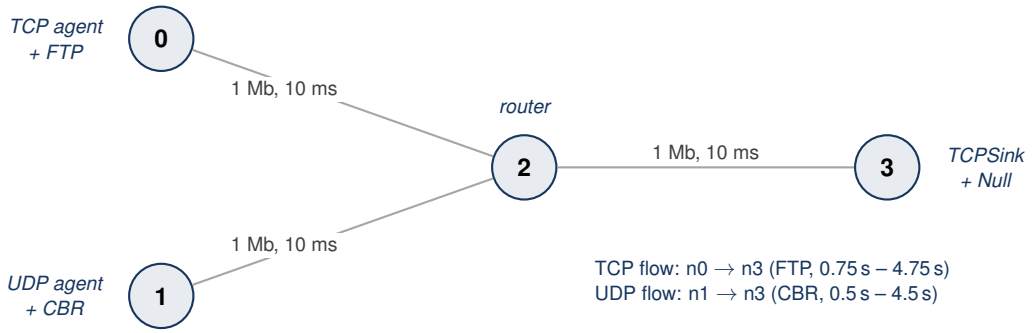


Figure 4. Four-node topology of Experiment 2. All links are 1 Mb, 10 ms, DropTail with a queue limit of 50 packets.

Procedure

1. Create the simulator, the NAM and trace files and the `finish` procedure.
2. Create four nodes, label them and join n0–n2, n1–n2 and n2–n3 with duplex links (1 Mb, 10 ms, DropTail); set the orientation and a queue limit of 50 packets.
3. Create a TCP agent on n0 and a TCPSink on n3, connect them and attach an FTP application to the TCP agent.
4. Create a UDP agent on n1 and a Null agent on n3, connect them and attach a CBR application (500 bytes, 0.005 s interval).
5. Start the CBR traffic at 0.5 s and stop it at 4.5 s; start the FTP traffic at 0.75 s and stop it at 4.75 s; call `finish` at 5.0 s.
6. Run `ns exp2.tcl`; count the TCP and UDP records with `exp2.awk`.
7. Repeat for other link bandwidths and record the counts in the files `plot_tcp` and `plot_udp`; plot them with XGraph.
8. Generate the throughput data with `exp2_throughput.awk` and plot it against time.

Program

`exp2.tcl` — NS-2 simulation script

```
set ns [new Simulator]

#open a new file for NAMTRACE
set nf [open out.nam w]
$ns namtrace-all $nf

#open a new file to log TRACE
set tf [open out.tr w]
$ns trace-all $tf

#body of finish procedure
proc finish {} {
    global ns nf tf
    $ns flush-trace
    close $nf
    close $tf
    exec nam out.nam &
    exit 0
```

```
}

#Create 4 nodes
set n0 [$ns node]
set n1 [$ns node]
set n2 [$ns node]
set n3 [$ns node]

#set label
$n0 label "TCP"
$n1 label "UDP"
$n2 label "Receiver"

#Create links between nodes
$ns duplex-link $n0 $n2 1Mb 10ms DropTail
$ns duplex-link $n1 $n2 1Mb 10ms DropTail
$ns duplex-link $n2 $n3 1Mb 10ms DropTail

#set orientation
$ns duplex-link-op $n0 $n2 orient right-down
$ns duplex-link-op $n1 $n2 orient right-up
$ns duplex-link-op $n2 $n3 orient right

#set the queue limit-default is 50 packets
$ns queue-limit $n0 $n2 50
$ns queue-limit $n1 $n2 50
$ns queue-limit $n2 $n3 50

#Create TCP agent b/w node 0 and node 3
#Sender
set tcp0 [new Agent/TCP]
$ns attach-agent $n0 $tcp0

#Receiver
set sink0 [new Agent/TCPSink]
$ns attach-agent $n3 $sink0
$ns connect $tcp0 $sink0

#Create FTP Application for TCP Agent
set ftp0 [new Application/FTP]
$ftp0 attach-agent $tcp0

#Specify TCP packet size
Agent/TCP set packetSize_ 1000

#Create UDP Agent b/w node1 and node3
#Sender
set udp0 [new Agent/UDP]
$ns attach-agent $n1 $udp0

#Receiver
set null0 [new Agent/Null]
$ns attach-agent $n3 $null0
$ns connect $udp0 $null0

#Create CBR Application for UDP Agent
set cbr0 [new Application/Traffic/CBR]
$cbr0 set packetSize_ 500
$cbr0 set interval_ 0.005
$cbr0 attach-agent $udp0

#Start and Stop FTP Traffic
$ns at 0.75 "$ftp0 start"
$ns at 4.75 "$ftp0 stop"

#Start and stop CBR traffic
$ns at 0.5 "$cbr0 start"
$ns at 4.5 "$cbr0 stop"

#Stop the simulations
$ns at 5.0 "finish"
```

```
$ns run
```

exp2.awk — packet counts of TCP and UDP

```
#gedit exp2.awk
BEGIN {
    tcp_count = 0;
    udp_count = 0;
}
{
    if($5 == "cbr")
        udp_count++;
    else if($5 == "tcp")
        tcp_count++;
}
END {
    printf("The number of packets by TCP =%d\n", tcp_count);
    printf("The number of packets by UDP =%d\n", udp_count);
}
```

exp2_throughput.awk — throughput against time

```
#gedit exp2_throughput.awk
BEGIN {
    sSize = 0;
    startTime = 5.0;
    stopTime = 0.1;
    Tput = 0;
}
{
    event = $1;
    time = $2;
    size = $6;
    if(event == "+")
    {
        if (time < startTime) {
            startTime = time;
        }
    }
    if(event == "r")
    {
        if(time > stopTime)
        {
            stopTime = time;
        }
        sSize += size;
    }
    Tput = (sSize / (stopTime - startTime)) * (8/1000);
    printf("%f\t %.2f\n", time, Tput);
}
END {
}
```

Execution and output

Terminal

```
$ ns exp2.tcl
$ awk -f exp2.awk out.tr
The number of packets by TCP =846
The number of packets by UDP =4786
```

The throughput data is written to a file and plotted; the file has one `time throughput` pair per line. The first values are (negative) zero because nothing has been received yet.

Terminal

```
$ awk -f exp2_throughput.awk out.tr > plot_throughput
$ xgraph plot_throughput
$ head -3 plot_throughput; tail -2 plot_throughput
0.500000          -0.00
0.500000          -0.00
0.505000          -0.00
...
4.800920          2017.06
4.809240          2013.24
```

Observations

The simulation was repeated for five link bandwidths. The counts are the values stored in `plot_tcp` (TCP) and `plot_udp` (UDP).

Bandwidth (Mb)	TCP records	UDP records
1	846	4786
1.25	1509	4791
1.5	2071	4805
1.75	2892	4806
2	3582	4806

Table 3: number of TCP and UDP packet records against link bandwidth

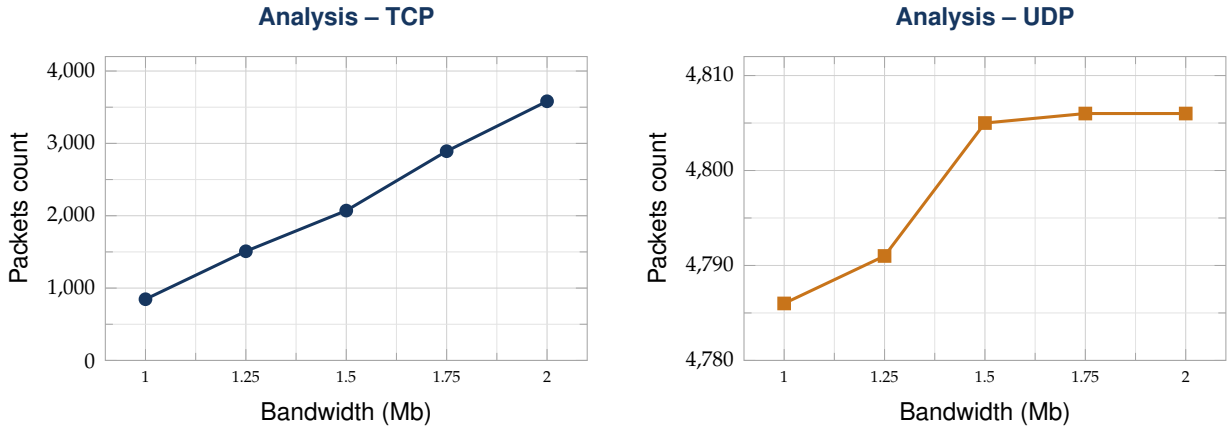


Figure 5. TCP and UDP packet counts for different link bandwidths (`xgraph plot_tcp` and `xgraph plot_udp`). Note the very different vertical scales.

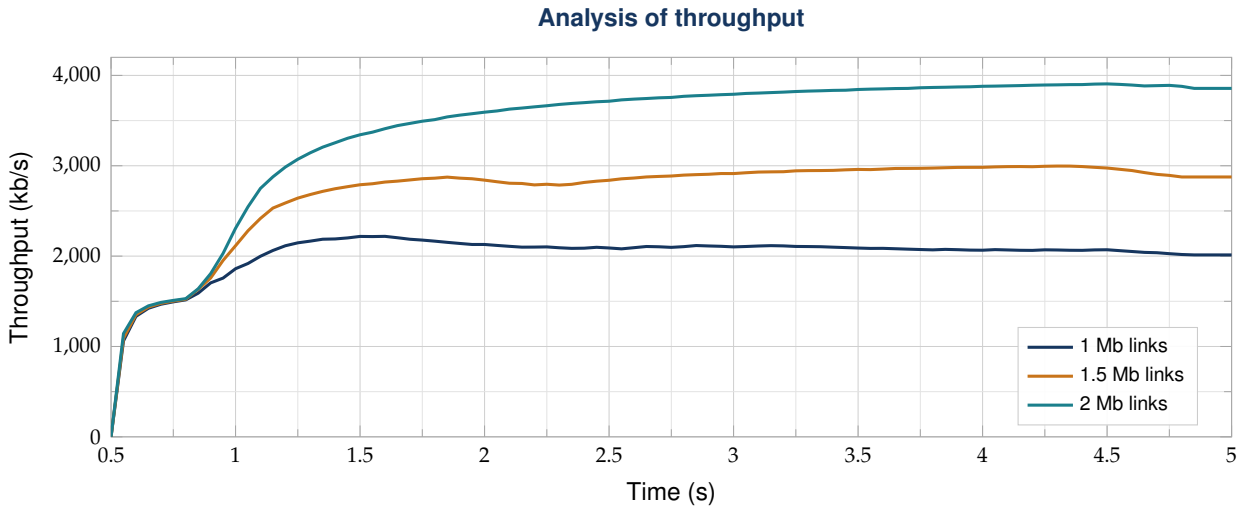


Figure 6. Aggregate throughput against time (`xgraph plot_throughput`) for three link bandwidths; each curve was produced by re-running `exp2.tcl` with the bandwidth changed.

- The **TCP** count grows from 846 to 3582 (about four times) when the bandwidth is doubled from 1 Mb to 2 Mb, because TCP uses the capacity that the UDP flow leaves free.
- The **UDP** count stays almost constant (4786 to 4806): the CBR source sends at a fixed rate irrespective of the available bandwidth.
- The throughput rises quickly once the sources start at 0.5 s and then levels off. At the end of the run it is about 2013 kb/s (1 Mb), 2876 kb/s (1.5 Mb) and 3856 kb/s (2 Mb): a higher bandwidth gives a higher throughput.

Result. A four-node network carrying a TCP flow (n0 → n3) and a UDP flow (n1 → n3) was simulated in NS-2. The TCP traffic increased with the link bandwidth (846 to 3582 records for 1 Mb to 2 Mb), the UDP traffic remained nearly constant, and the network throughput increased with the bandwidth.

Viva voce questions for this experiment: see Appendix A, page 30.

EXPERIMENT NO. 3 Date: 28/08/2026

Ethernet LAN: Throughput versus Data Rate and Error Rate

Aim

> Simulate an Ethernet LAN using N nodes (6–10). Change the error rate and the data rate and compare the throughput.

Theory

Ethernet and CSMA/CD

In an Ethernet LAN (IEEE 802.3) all stations share one transmission medium. A station first *senses* the channel and transmits only when it is idle (carrier sense, multiple access). While transmitting it keeps listening; if another station transmits at the same time a *collision* is detected, both stations stop, and each waits for a random time before trying again. The waiting time follows *binary exponential backoff*: after the i-th collision a station waits a random number of slot times between 0 and $2^{\min(i,10)} - 1$, and gives up after 16 attempts.

LANs in NS-2

The command

```
$ns make-lan "$n1 $n2 $n3 $n4 $n5 $n6" 2Mb 40ms LL Queue/DropTail Mac/802_3
```

creates a shared LAN of the listed nodes with a bandwidth of 2 Mb/s and a delay of 40 ms. Every node gets a link layer (`LL`), an interface queue (`Queue/DropTail`) and an Ethernet MAC (`Mac/802_3`) that implements CSMA/CD. In this experiment seven nodes are used: n0 is joined to the LAN node n1 by a point-to-point link (3 Mb, 20 ms), and n1–n6 form the LAN. A TCP source with an FTP application on n0 sends to a TCP sink on n6.

Data rate and error rate

The *data rate* is the bandwidth given to `make-lan`. An *error rate* is introduced with an NS-2 `ErrorModel` object that discards each packet crossing a link with a given probability. TCP treats every loss as a sign of congestion and reduces its window (or waits for a timeout), so both a lower data rate and a higher error rate reduce the throughput. The throughput is computed by `exp3.awk` with the same formula as in Experiment 2.

Network topology

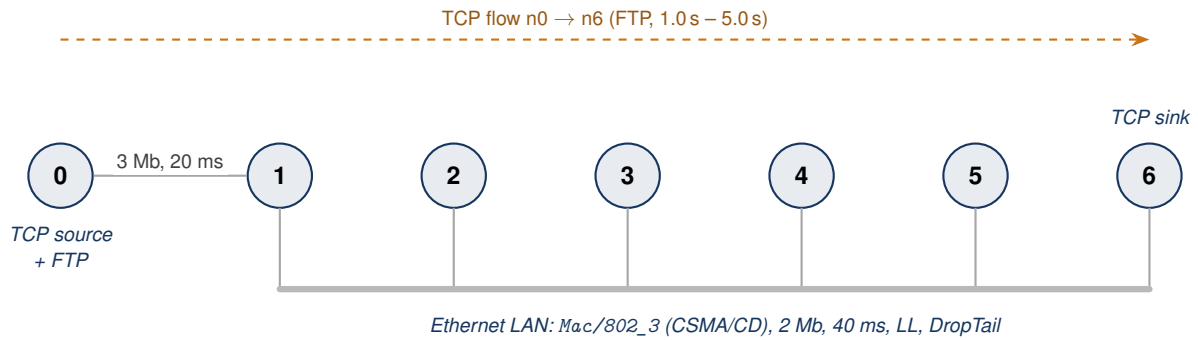


Figure 7. Topology of Experiment 3: a point-to-point link followed by a six-node Ethernet LAN.

Procedure

1. Create the simulator, the NAM file `exp3.nam`, the trace file `exp3.tr` and a `finish` procedure that also runs `exp3.awk` to produce `exp3.graph`.
2. Create seven nodes, label n0 (TCP source) and n6 (TCP sink), and set the colours of the flow and of the nodes.
3. Join n0 and n1 with a 3 Mb, 20 ms duplex link and build the LAN of n1–n6 with `make-lan` (2 Mb, 40 ms, `Mac/802_3`).
4. Create a TCP agent on n0 and a TCPSink on n6, connect them, and attach an FTP application.
5. Start the FTP traffic at 1.0 s, stop it at 5.0 s and call `finish` at 5.5 s.
6. Run `ns exp3.tcl` and plot `exp3.graph` with XGraph.
7. **Change the data rate:** edit the LAN bandwidth (`2Mb`) and run again; keep a copy of `exp3.graph` for every rate and plot them together.
8. **Change the error rate:** insert the `ErrorModel` lines given below, set `rate_` to the required value, run again and compare the throughput.

Program

exp3.tcl — NS-2 simulation script

```
set ns [new Simulator]
set namfile [open exp3.nam w]
$ns namtrace-all $namfile
set tracefile [open exp3.tr w]
$ns trace-all $tracefile

proc finish {} {
    global ns namfile tracefile
    $ns flush-trace
    close $namfile
    close $tracefile
    exec nam exp3.nam &
    exec awk -f exp3.awk exp3.tr > exp3.graph &
    exit 0
}

#Create 6 nodes
```

```
set n0 [$ns node]
set n1 [$ns node]
set n2 [$ns node]
set n3 [$ns node]
set n4 [$ns node]
set n5 [$ns node]
set n6 [$ns node]

#label source and sink nodes
$n0 label "TCPSource"
$n6 label "TCPSink"

#color to the flow
$ns color 1 red

#color to the nodes
$n0 color blue
$n6 color orange

$ns duplex-link $n0 $n1 3Mb 20ms DropTail
$ns make-lan "$n1 $n2 $n3 $n4 $n5 $n6" 2Mb 40ms LL Queue/DropTail Mac/802_3

$ns duplex-link-op $n0 $n1 orient right
set tcp [new Agent/TCP]
$ns attach-agent $n0 $tcp
$tcp set class_ 1

set tcpsink [new Agent/TCPSink]
$ns attach-agent $n6 $tcpsink
$ns connect $tcp $tcpsink

set ftp [new Application/FTP]
$ftp attach-agent $tcp

$ns at 1.0 "$ftp start"
$ns at 5.0 "$ftp stop"
$ns at 5.5 "finish"
$ns run
```

exp3.awk — throughput against time

```
#gedit exp3.awk
BEGIN {
    sSize = 0;
    startTime = 5.0;
    stopTime = 0.1;
    Tput = 0;
}
{
    event = $1;
    time = $2;
    size = $6;
    if (event == "+")
    {
        if (time < startTime)
        {
            startTime = time;
        }
    }
    if (event == "r")
    {
        if (time > stopTime)
        {
            stopTime = time;
        }
        sSize += size;
    }
    Tput = (sSize/(stopTime-startTime)) * (8/1000);
    printf("%f\t %.2f\n", time, Tput);
}
END {
```

```
}
```

Lines added to introduce an error rate

The following lines are inserted into `exp3.tcl` directly after the command that creates the n0–n1 link. `rate_ 0.01` drops 1 % of the packets that cross the link from n0 to n1.

error model (added after the n0-n1 duplex-link command)

```
#Error model on the link n0 -> n1
set em [new ErrorModel]
$em unit pkt
$em set rate_ 0.01
$em ranvar [new RandomVariable/Uniform]
$em drop-target [new Agent/Null]
$ns lossmodel $em $n0 $n1
```

Execution and output

Terminal

```
$ ns exp3.tcl
$ xgraph exp3.graph
$ head -3 exp3.graph; tail -2 exp3.graph
1.000000          -0.00
1.000000          -0.00
1.020107          15.91
...
5.202373          1328.39
5.206637          1327.12
```

`exp3.graph` is written by the `awk` command inside the `finish` procedure. When the `ns` program prints `warning: no class variable LanRouter::debug_` this is normal for `make-lan` and can be ignored.

Observations

Effect of the data rate

The LAN bandwidth was changed from 2 Mb to 0.5 Mb, 1 Mb and 3 Mb while everything else was kept constant.

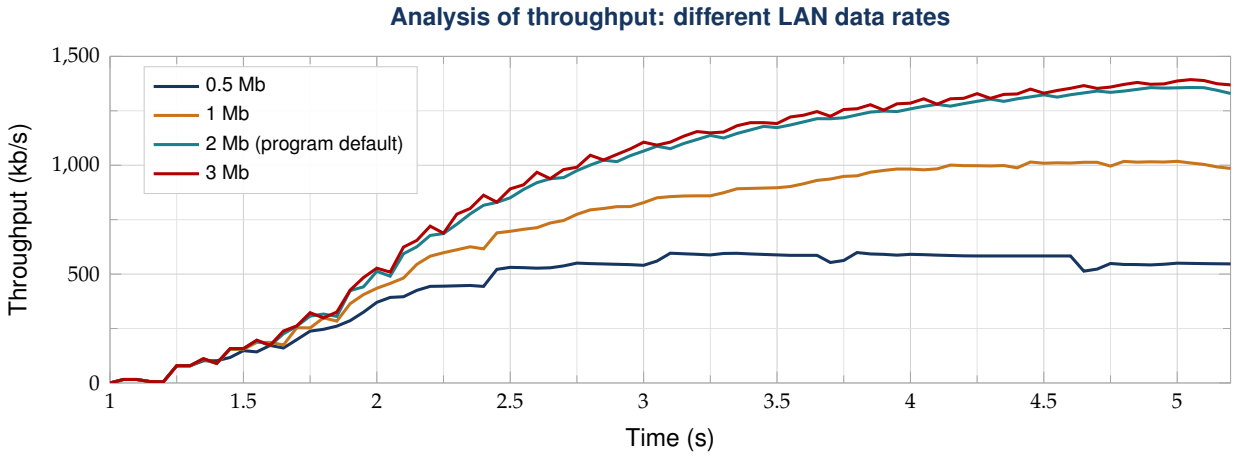


Figure 8. Throughput against time for four LAN data rates (`xgraph exp3.graph` for each run).

Effect of the error rate

The `ErrorModel` of the previous page was inserted with `rate_` set to 0, 0.001, 0.005 and 0.01 (error rates of 0 %, 0.1 %, 0.5 % and 1 %); the LAN data rate was 2 Mb.

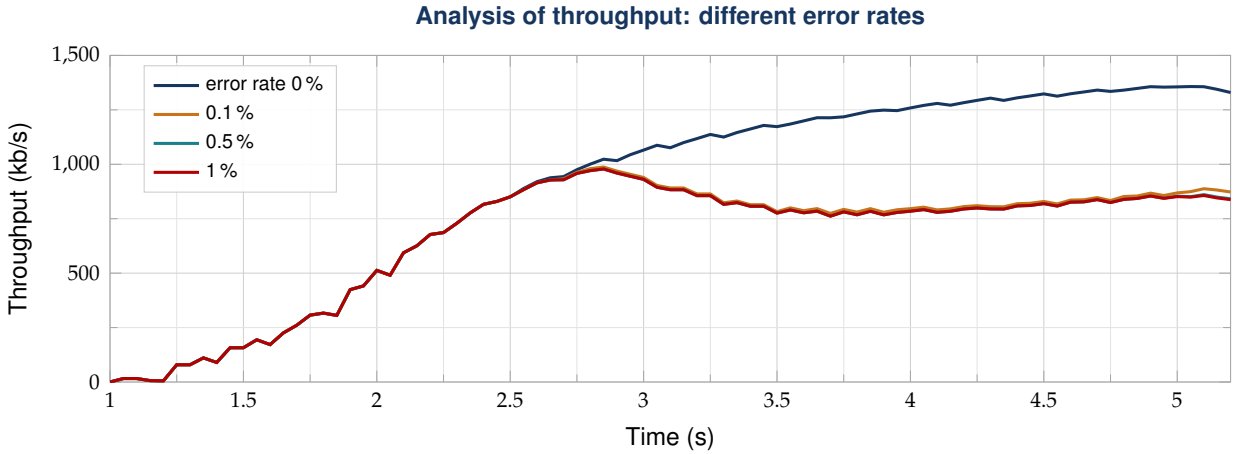


Figure 9. Throughput against time for four error rates on the link n0–n1.

LAN data rate	Throughput at 5.2 s
0.5 Mb	535 kb/s
1 Mb	985 kb/s
2 Mb	1327 kb/s
3 Mb	1369 kb/s

Table 4: error rate = 0

Error rate	Throughput at 5.2 s
0 %	1327 kb/s
0.1 %	871 kb/s
0.5 %	842 kb/s
1 %	805 kb/s

Table 5: LAN data rate = 2 Mb

- **Data rate.** The throughput rises almost in proportion to the data rate from 0.5 Mb to 1 Mb (535 to 985 kb/s). Beyond 2 Mb the gain is small (1327 to 1369 kb/s): the LAN is then no longer the bottleneck and the growth of the TCP window over a path with a round-trip delay of at least 120 ms limits the rate.
- **Error rate.** Even a small error rate lowers the throughput clearly, from 1327 kb/s without errors to 871 kb/s at 0.1 % and 805 kb/s at 1 %, because each lost packet makes TCP reduce its window or wait for a retransmission timeout. The exact values depend on which packets are lost, since the error model drops packets at random.

Result. An Ethernet LAN of six nodes (n1–n6) with a TCP source on n0 and a sink on n6 was simulated in NS-2. The throughput increased with the data rate until the TCP flow itself became the limit (535 kb/s at 0.5 Mb to 1369 kb/s at 3 Mb), and it decreased when an error rate was introduced (1327 kb/s without errors, 805 kb/s at 1 % errors).

Viva voce questions for this experiment: see Appendix A, page 30.

EXPERIMENT NO. 4 Date: 04/09/2026

Ethernet LAN with Multiple Traffic Flows: Congestion Window of TCP Vegas and Reno

Aim

> Simulate an Ethernet LAN using N nodes and set multiple traffic nodes, and determine the collisions across the different nodes. Also plot the congestion window for different source/destination pairs.

Theory

Congestion window

A TCP sender may have only a limited amount of unacknowledged data in the network. This limit is the *congestion window* (`cwnd_`), measured here in packets. In *slow start* the window grows by one packet per acknowledgement, i.e. it doubles every round-trip time; in *congestion avoidance* it grows by about one packet per round-trip time. In NS-2 the number of packets actually in flight is also bounded by the window parameter `window_` (20 packets by default), while the variable `cwnd_` itself keeps changing according to the TCP variant.

TCP Reno and TCP Vegas

Reno is loss based: it keeps increasing the window until a packet is lost and then halves it (fast retransmit and fast recovery). **Vegas** is delay based: it compares the expected rate (window divided by the smallest round-trip time) with the actual rate and keeps the difference between two thresholds, so that only a few packets are queued in the network. The window therefore settles at a steady value instead of growing until a loss occurs.

Shared LAN and collisions

Two TCP flows, n1 $\rightarrow$ n7 (Vegas) and n2 $\rightarrow$ n8 (Reno), cross the router n0 and then the Ethernet LAN formed by n3–n8 (512 kb/s, 40 ms, `Mac/802_3`). On the shared LAN the data packets and the acknowledgements of both flows compete for the medium; CSMA/CD detects collisions and resolves them by random backoff, as described in Experiment 3. Colours and shapes (`color`, `shape`) are used only to make the flows and the nodes n7 (box) and n8 (hexagon) recognisable in NAM. The congestion window of a connection is written to a file with `$tcp1 attach $tfile1` and `$tcp1 trace cwnd_`.

Network topology

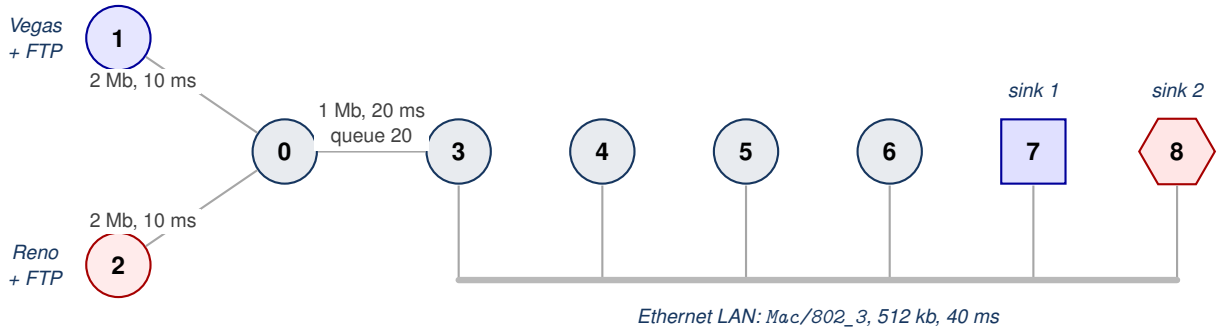


Figure 10. Topology of Experiment 4. Flow 1 (blue): TCP Vegas n1 → n7; flow 2 (red): TCP Reno n2 → n8.

Flow	TCP variant	Path	Starts	`class_`	cwnd file	Plot file
1	Vegas (`Agent/TCP/Vegas`)	n1 → n7	0.5 s	1 (blue)	`cwnd1.tr`	`tcp1`
2	Reno (`Agent/TCP/Reno`)	n2 → n8	1.0 s	2 (red)	`cwnd2.tr`	`tcp2`

Table 6: the two traffic flows of Experiment 4 (both stop at 5.0 s; packet size 55 bytes)

Procedure

1. Create the simulator, the NAM file `exp4.nam`, the trace file `exp4.tr`, and a `finish` procedure that closes the congestion-window files and extracts `tcp1` and `tcp2` with `exp4.awk`.
2. Create nine nodes n0–n8; set the flow colours (1 blue, 2 red) and the shapes and colours of n7 and n8.
3. Join n1–n0 and n2–n0 (2 Mb, 10 ms) and n0–n3 (1 Mb, 20 ms), build the LAN n3–n8 with `make-lan` (512 Kb, 40 ms, `Mac/802_3`) and set the queue limit of n0–n3 to 20 packets.
4. Create the TCP Vegas agent on n1 with its sink on n7 and an FTP source; create the TCP Reno agent on n2 with its sink on n8 and an FTP source.
5. Open `cwnd1.tr` and `cwnd2.tr`, attach them to the agents and trace `cwnd_`.
6. Start FTP 1 at 0.5 s and FTP 2 at 1.0 s, stop both at 5.0 s and call `finish` at 5.5 s.
7. Run `ns exp4.tcl` and plot the two files with `xgraph tcp1` and `xgraph tcp2`.

Program

`exp4.tcl` — NS-2 simulation script

```
set ns [new Simulator]
set namfile [open exp4.nam w]
$ns namtrace-all $namfile

set tracefile [open exp4.tr w]
$ns trace-all $tracefile

proc finish {} {
    global ns namfile tracefile tfile1 tfile2
    $ns flush-trace
    close $namfile
    close $tracefile
    close $tfile1
```

```
    close $tfile2
    exec nam exp4.nam &
    exec awk -f exp4.awk cwnd1.tr > tcp1 &
    exec awk -f exp4.awk cwnd2.tr > tcp2 &
    exit 0
}

#Create the nodes
set n0 [$ns node]
set n1 [$ns node]
set n2 [$ns node]
set n3 [$ns node]
set n4 [$ns node]
set n5 [$ns node]
set n6 [$ns node]
set n7 [$ns node]
set n8 [$ns node]

#color to packet flow
$ns color 1 Blue
$ns color 2 Red

#set shape and color to node 7 and node 8
$n7 shape Box
$n7 color Blue
$n8 shape hexagon
$n8 color Red

$ns duplex-link $n1 $n0 2Mb 10ms DropTail
$ns duplex-link $n2 $n0 2Mb 10ms DropTail
$ns duplex-link $n0 $n3 1Mb 20ms DropTail

#Create LAN
$ns make-lan "$n3 $n4 $n5 $n6 $n7 $n8" 512Kb 40ms LL Queue/DropTail Mac/802_3

#Set Orientation
$ns duplex-link-op $n1 $n0 orient right-down
$ns duplex-link-op $n2 $n0 orient right-up
$ns duplex-link-op $n0 $n3 orient right
$ns queue-limit $n0 $n3 20

#set the transport protocol
set tcp1 [new Agent/TCP/Vegas]
$ns attach-agent $n1 $tcp1

set sink1 [new Agent/TCPSink]
$ns attach-agent $n7 $sink1

#set the application
set ftp1 [new Application/FTP]
$ftp1 attach-agent $tcp1

$ns connect $tcp1 $sink1
$tcp1 set class_ 1
$tcp1 set packetSize_ 55

set tfile1 [open cwnd1.tr w]
$tcp1 attach $tfile1
$tcp1 trace cwnd_

set tcp2 [new Agent/TCP/Reno]
$ns attach-agent $n2 $tcp2

set sink2 [new Agent/TCPSink]
$ns attach-agent $n8 $sink2

set ftp2 [new Application/FTP]
$ftp2 attach-agent $tcp2

$ns connect $tcp2 $sink2
$tcp2 set class_ 2
```

```
$tcp2 set packetSize_ 55

set tfile2 [open cwnd2.tr w]
$tcp2 attach $tfile2
$tcp2 trace cwnd_

$ns at 0.5 "$ftp1 start"
$ns at 1.0 "$ftp2 start"
$ns at 5.0 "$ftp2 stop"
$ns at 5.0 "$ftp1 stop"
$ns at 5.5 "finish"
$ns run
```

exp4.awk — extracts time and congestion window

```
# exp4.awk
BEGIN {
}
{
    if($6 == "cwnd_")
    {
        printf("%f\t %f\n", $1, $7);
    }
}
END {
}
```

Execution and output

Terminal

```
$ ns exp4.tcl
$ xgraph tcp1
$ xgraph tcp2
```

The files written by NS-2 contain one line per change of the congestion window: time, source node, source port, destination node, destination port, the traced variable and its value. The AWK script keeps lines whose sixth field is `cwnd_` and prints the time and the value.

cwnd1.tr (first line) and tcp1 (extracted data)

```
0.00000  1  0  7  0  cwnd_ 1.000
0.94645  1  0  7  0  cwnd_ 2.000
$ head -4 tcp1; tail -2 tcp1
0.000000          1.000000
0.946450          2.000000
0.946450          3.000000
1.392900          4.000000
...
5.483430          1.000000
5.483430          2.000000
```

Observations

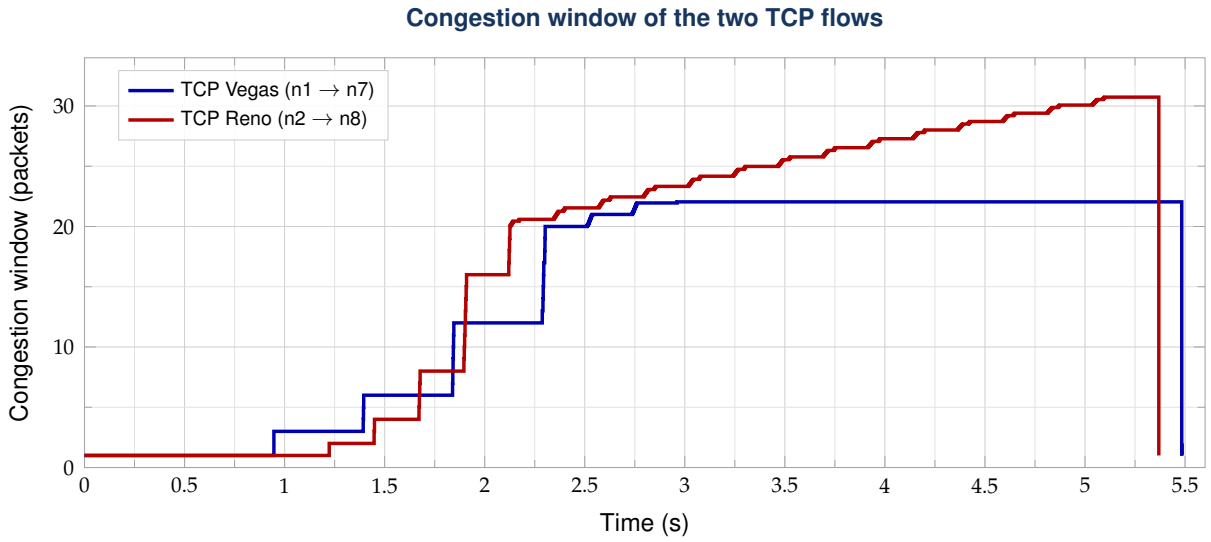


Figure 11. Congestion window against time (`xgraph tcp1` and `xgraph tcp2` drawn in one plot).

- Both connections begin in **slow start**: the window grows quickly from 1 packet (Vegas from 0.5 s, Reno from 1.0 s).
- **Vegas** leaves slow start at 20 packets (2.3 s, the NS-2 default threshold), grows slowly to 22.0 packets at $t = 2.96$ s and then keeps this value: no further change was logged until the sources stopped. Its delay-based control has found a steady operating point.
- **Reno** leaves slow start at about 2.1 s (also at 20 packets) and then grows linearly (congestion avoidance), reaching 30.7 packets at $t = 5.10$ s. It never reduces the window because no packet is lost: the trace file contains no `d` (drop) records.
- Both curves end with a reset of the window to 1 packet shortly after the FTP sources stop at 5.0 s.
- The two flows share the LAN without any dropped packet: contention for the medium is resolved by CSMA/CD backoff and every packet is delivered.

Result. An Ethernet LAN with two simultaneous TCP flows was simulated in NS-2, and the congestion windows of TCP Vegas (n1 → n7) and TCP Reno (n2 → n8) were extracted from the trace and plotted. Vegas settled at a constant window of about 22 packets, whereas Reno kept increasing its window (30.7 packets at 5.1 s); no packet was dropped.

Viva voce questions for this experiment: see Appendix A, page 31.

EXPERIMENT NO. 5 Date: 11/09/2026

Bit Stuffing in the Data-Link Layer

Aim

> Write a program to implement the bit stuffing (or byte stuffing) concept in the data-link layer.

Theory

Framing and the need for stuffing

The data-link layer groups the bits it receives from the network layer into *frames*. One way of marking the start and the end of a frame is a special *flag* pattern; HDLC uses the bit pattern `01111110`. If the data itself contained this pattern, the receiver would mistake it for the end of the frame. *Stuffing* makes the data transparent by making sure that the flag pattern can never occur inside the data.

Bit stuffing

- **Sender:** whenever five consecutive 1s have been sent, a 0 is inserted immediately after them, whatever the next data bit is.
- **Receiver:** whenever five consecutive 1s are received, the next bit (a stuffed 0) is removed.

Hence six consecutive 1s can only occur inside a flag. In the worst case, when the data consists only of 1s, one extra bit is added for every five data bits (an overhead of 20 %).

Byte stuffing

Character-oriented protocols (for example PPP) use a flag *byte* (`0x7E`) and an escape byte (`0x7D`). Every flag or escape byte inside the data is preceded by an escape byte (PPP additionally XORs the data byte with `0x20`). The program of this experiment implements **bit stuffing**.

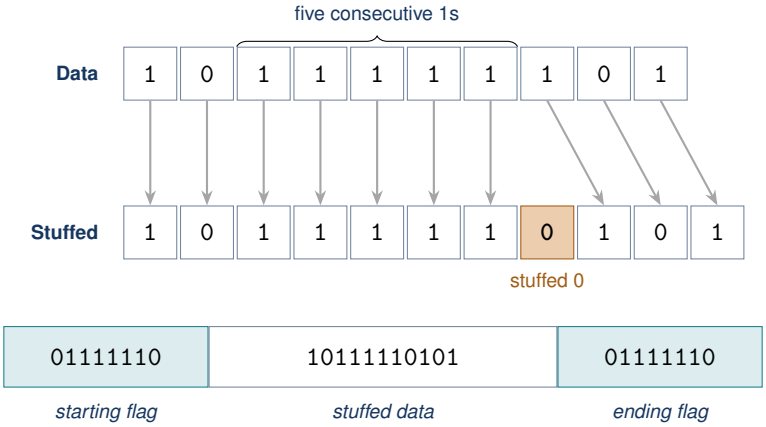


Figure 12. Bit stuffing of the data `1011111101` and the resulting frame (the stuffed 0 is highlighted).

Algorithm

Bit stuffing (sender)

1. Set `count = 0` and let `stuffed_data` be empty.
2. For each data bit, append the bit to `stuffed_data`.
3. If the bit is `1`, add 1 to `count`; when `count` reaches 5, append a `0` and set `count = 0`.
4. If the bit is `0`, set `count = 0`.
5. Return `stuffed_data`.

Bit destuffing (receiver)

1. Set `count = 0`, `skip_next = False` and let `destuffed_data` be empty.
2. For each received bit: if `skip_next` is set, clear it and ignore the bit (it is the stuffed 0).
3. Otherwise append the bit; if it is `1`, add 1 to `count`; when `count` reaches 5, set `skip_next` and `count = 0`.
4. If the bit is `0`, set `count = 0`.
5. Return `destuffed_data`.

Program

exp5.py — bit stuffing and destuffing

```
def bit_stuffing(data):
    stuffed_data = ""
    count = 0
    for bit in data:
        stuffed_data += bit
        if bit == '1':
            count += 1
            if count == 5:
                stuffed_data += '0'
                count = 0
        else:
            count = 0
    return stuffed_data

def bit_destuffing(stuffed_data):
    destuffed_data = ""
    count = 0
    skip_next = False
    for bit in stuffed_data:
        if skip_next:
            skip_next = False
            continue
        destuffed_data += bit
        if bit == '1':
            count += 1
            if count == 5:
                skip_next = True
                count = 0
        else:
            count = 0
    return destuffed_data

data = input("Enter the input data:")
print("Original data:", data)
stuffed = bit_stuffing(data)
print("Bit Stuffed data:", stuffed)
destuffed = bit_destuffing(stuffed)
print("Bit Destuffed data:", destuffed)
```

Execution and output

Terminal

```
$ python3 exp5.py
Enter the input data:1011111101
Original data: 1011111101
Bit Stuffed data: 10111110101
Bit Destuffed data: 1011111101
```

Further test runs, including the special cases of five and ten consecutive 1s and of data without any run of 1s:

Input data	Bit stuffed data	Bit destuffed data
1011111101	10111110101	1011111101
11111	111110	11111
1111111111	111110111110	1111111111
0000	0000	0000

Table 7: results of the program for four input strings

Observations

- A `0` is inserted after every run of five consecutive 1s, even when the run ends the data (`11111` → `111110`); data without such a run is unchanged (`0000`).
- The destuffed data is identical to the original data in every case, so the transmission is transparent.
- For the all-ones input of ten bits two extra bits are added, which is the 20 % worst-case overhead.

Result. A Python program was written to perform bit stuffing at the sender and bit destuffing at the receiver. The input `1011111101` was stuffed to `10111110101` and destuffed back to the original data.

Viva voce questions for this experiment: see Appendix A, page 31.

EXPERIMENT NO. 6 Date: 09/10/2026

Error Detection Using CRC (CRC-CCITT)

Aim

> Write a program for an error-detecting code using CRC-CCITT (16-bit).

Theory

Cyclic redundancy check

A CRC treats a bit string as a polynomial with binary coefficients; for example `1001` is $x^3 + 1$. Sender and receiver agree on a *generator polynomial* $G(x)$ of degree r (that is, $r + 1$ bits). All arithmetic is modulo 2: addition and subtraction are both the exclusive OR (XOR) and there are no carries.

1. **Sender:** append r zeros to the k message bits, giving $M(x)\,x^r$.
2. Divide $M(x)\,x^r$ by $G(x)$ (modulo 2) and keep the r-bit remainder $R(x)$.
3. Transmit $T(x) = M(x)\,x^r + R(x)$; this codeword is exactly divisible by $G(x)$.
4. **Receiver:** divide the received bits by $G(x)$. A remainder of zero means that no error was detected; a non-zero remainder means that the frame is corrupted.

CRC-CCITT

The CRC-CCITT generator is $G(x) = x^{16} + x^{12} + x^5 + 1$, i.e. the 17-bit pattern `1 0001 0000 0010 0001` (hexadecimal `0x11021`), so $r = 16$ check bits are appended. A 16-bit CRC detects all single-bit and double-bit errors, all errors with an odd number of bits, all burst errors of length 16 or less, and 99.997 % of 17-bit bursts. An error pattern goes undetected only if it is itself divisible by $G(x)$.

The program of this experiment keeps the generator in the list `div`. To keep the hand calculation short it is set to `[1, 0, 1, 1]`, i.e. $G(x) = x^3 + x + 1$ and $r = 3$; the program works unchanged for any generator, and the replacement that gives the 16-bit CRC-CCITT is shown at the end of this experiment.

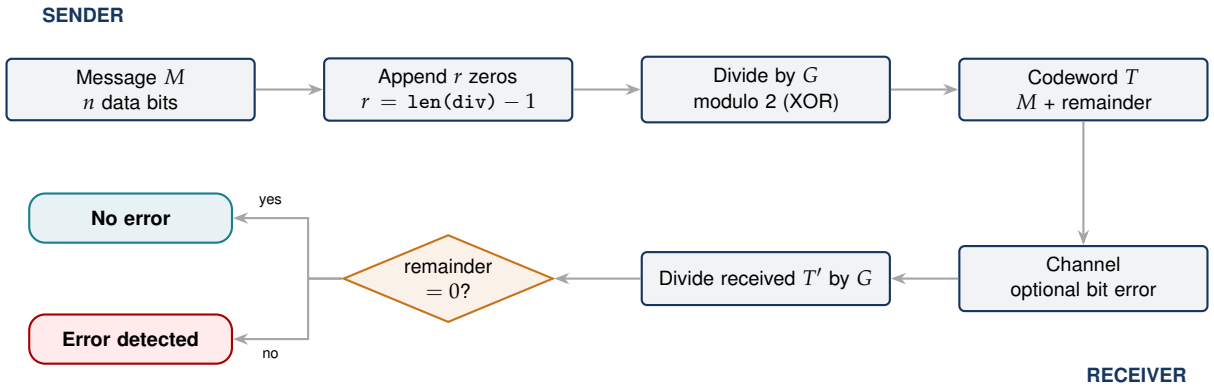


Figure 13. Principle of CRC error detection.

Worked example

Message `1001` and generator `1011` ($r = 3$). Three zeros are appended and the division is carried out by XOR-ing the generator under every leading 1:

Sender: 1001000 divided by 1011

```
       1 0 0 1 0 0 0
XOR    1 0 1 1
       -------------
       0 0 1 0 0 0 0
XOR        1 0 1 1
       -------------
       0 0 0 0 1 1 0      remainder = 110   ->   codeword = 1001 110
```

The receiver divides the received codeword `1001110` in the same way and obtains the remainder `000`. If the first bit is corrupted (`0001110`) the remainder is `101`:

Receiver: error-free codeword and corrupted codeword

```
received 1001110                          received 0001110 (bit 1 flipped)
       1 0 0 1 1 1 0                             0 0 0 1 1 1 0
XOR    1 0 1 1                                   (bits 0-2 are 0: no XOR)
       -------------                      XOR          1 0 1 1
       0 0 1 0 1 1 0                             -------------
XOR        1 0 1 1                               0 0 0 0 1 0 1
       -------------
       0 0 0 0 0 0 0      remainder 000          remainder 101 -> error
```

Algorithm

1. Read the number of data bits n and the bits; let m be the length of the generator `div`. If $n < m$ the data is invalid.
2. Copy the data to `cdata` and append $m - 1$ zeros (initial codeword).
3. For $i = 0, \ldots, n - 1$: if `cdata[i]` is 1, XOR the m generator bits into `cdata[i..i+m-1]`.
4. The last $m - 1$ bits of `cdata` are the remainder; append them to the data to form the codeword and display it.
5. Optionally flip the bit at a position chosen by the user (error introduction) and display the received message.
6. Divide the received message by the generator again; if every bit of the result is 0 print *No Error*, otherwise print *Error!*.

Program

exp6.py — CRC generation and error detection

```python
div = [1, 0, 1, 1]
cdata = []
n = int(input("Enter the number of bits:"))
data = [int(input(f"Enter bit {i + 1}:")) for i in range(n)]
m = len(div)

if n < m:
    print("Invalid data")
else:
    cdata = data.copy()
    for i in range(m - 1):
        cdata.append(0)
    print("Initial Codeword:", cdata)
```

```
    for i in range(n):
        if cdata[i] == 1:
            for j in range(m):
                cdata[i + j] ^= div[j]

    for i in range(n, n + m - 1):
        data.append(cdata[i])
    print("Codeword to be transmitted is:", data)
    ch = input("Introduce Error - Y -> YES N->NO:")
    if ch.upper() == 'Y':
        p = int(input("Enter the position to change bit:"))
        if 1 <= p <= len(data):
            data[p-1] ^= 1
        else:
            print("Invalid Position!")
    print("The message received is:", data)
    for i in range(n):
        if data[i] == 1:
            for j in range(m):
                data[i + j] ^= div[j]
    flag = True
    for i in data:
        if i == 1:
            flag = False
            break
    if (flag):
        print("No Error\nSuccessfully Transferred")
    else:
        print("Error!")
```

Execution and output

Run 1: no error introduced

Terminal

```
$ python3 exp6.py
Enter the number of bits:4
Enter bit 1:1
Enter bit 2:0
Enter bit 3:0
Enter bit 4:1
Initial Codeword: [1, 0, 0, 1, 0, 0, 0]
Codeword to be transmitted is: [1, 0, 0, 1, 1, 1, 0]
Introduce Error - Y -> YES N->NO:N
The message received is: [1, 0, 0, 1, 1, 1, 0]
No Error
Successfully Transferred
```

Run 2: error introduced at position 1

Terminal

```
$ python3 exp6.py
Enter the number of bits:4
Enter bit 1:1
Enter bit 2:0
Enter bit 3:0
Enter bit 4:1
Initial Codeword: [1, 0, 0, 1, 0, 0, 0]
Codeword to be transmitted is: [1, 0, 0, 1, 1, 1, 0]
Introduce Error - Y -> YES N->NO:Y
Enter the position to change bit:1
The message received is: [0, 0, 0, 1, 1, 1, 0]
Error!
```

The initial codeword `[1, 0, 0, 1, 0, 0, 0]` is the data `1001` followed by three zeros; after the division the transmitted codeword is `[1, 0, 0, 1, 1, 1, 0]`, i.e. the data plus the CRC `110`, exactly as in the worked example.

The 16-bit CRC-CCITT generator

To obtain CRC-CCITT only the generator has to be replaced (the data must then contain at least 17 bits). The same program, run with 20 data bits `10110100111001011001`, gives:

exp6.py (first line replaced)

```
div = [1, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 1]      # x^16 + x^12 + x^5 + 1
```

CRC-CCITT, no error

```
$ python3 exp6.py
Enter the number of bits:20
...
Enter bit 20:1
...
Codeword to be transmitted is: [1, 0, 1, 1, 0, 1, 0, 0, 1, 1, 1, 0, 0, 1, 0, 1, 1, 0, 0,
1, 0, 0, 0, 1, 0, 1, 0, 1, 1, 1, 1, 0, 1, 1, 1, 0]
Introduce Error - Y -> YES N->NO:N
The message received is: [1, 0, 1, 1, 0, 1, 0, 0, 1, 1, 1, 0, 0, 1, 0, 1, 1, 0, 0, 1, 0,
0, 0, 1, 0, 1, 0, 1, 1, 1, 1, 0, 1, 1, 1, 0]
No Error
Successfully Transferred
```

CRC-CCITT, bit 5 flipped

```
$ python3 exp6.py
Enter the number of bits:20
...
Enter bit 20:1
...
Codeword to be transmitted is: [1, 0, 1, 1, 0, 1, 0, 0, 1, 1, 1, 0, 0, 1, 0, 1, 1, 0, 0,
1, 0, 0, 0, 1, 0, 1, 0, 1, 1, 1, 1, 0, 1, 1, 1, 0]
Introduce Error - Y -> YES N->NO:Y
Enter the position to change bit:5
The message received is: [1, 0, 1, 1, 1, 1, 0, 0, 1, 1, 1, 0, 0, 1, 0, 1, 1, 0, 0, 1, 0,
0, 0, 1, 0, 1, 0, 1, 1, 1, 1, 0, 1, 1, 1, 0]
Error!
```

Observations

- Without an error the remainder at the receiver is all zeros and the program prints *No Error / Successfully Transferred*.
- After a single bit of the codeword is flipped the remainder is non-zero and the program prints *Error!*; the same happens with the 16-bit generator.
- The CRC is appended as r extra bits (3 bits for `1011`, 16 bits for CRC-CCITT), whatever the length of the data.

Result. A Python program was written that appends a CRC to the data using modulo-2 division, lets the user corrupt one bit, and checks the received codeword. The data `1001` gave the codeword `1001110`; the error-free codeword was accepted and the codeword with the first bit flipped was reported as erroneous.

Viva voce questions for this experiment: see Appendix A, page 31.

APPENDIX A

Viva Voce Question Bank

Short questions with model answers for each experiment, for revision before the viva and the laboratory examination.

Experiment 1 — Point-to-Point Network: Queue Size, Bandwidth and Packet Drops

Q. What is NS-2 and which languages does it use?
Ans. NS-2 is a discrete-event network simulator. The simulation engine is written in C++ and the scenario is described in OTcl (Tcl) scripts.

Q. What do the event symbols `+`, `-`, `r` and `d` in the trace file mean?
Ans. Enqueue, dequeue, receive and drop respectively.

Q. Why is no packet dropped at 1 Mb but many at 0.25 Mb?
Ans. The CBR source offers $500 \times 8 / 0.005 = 800$ kb/s. A 1 Mb link can carry this; a 0.25 Mb link cannot, so its queue fills and DropTail discards the new arrivals.

Q. What is a DropTail queue and what is the default queue limit in NS-2?
Ans. A FIFO queue that drops the arriving packet when it is full; the default limit is 50 packets.

Q. Why does a larger queue reduce the number of drops?
Ans. The extra buffer space holds part of the excess traffic, so fewer packets arrive at a full queue (at the cost of a larger queuing delay).

Experiment 2 — TCP and UDP Traffic: Packet Counts and Throughput

Q. State the main differences between TCP and UDP.
Ans. TCP is connection-oriented and reliable with flow and congestion control; UDP is connectionless, unreliable and has no congestion control, so its rate is set only by the application.

Q. What are the roles of `Agent/TCPSink` and `Agent/Null`?
Ans. `TCPSink` receives TCP data and returns the acknowledgements; `Null` just absorbs UDP packets without any reply.

Q. What does `Application/FTP` do in NS-2?
Ans. It simulates a bulk file transfer, i.e. an unlimited supply of data for the TCP agent.

Q. Why does the UDP count hardly change with the bandwidth while the TCP count does?
Ans. The CBR source has a fixed rate (800 kb/s); TCP adapts its window to the capacity that remains.

Q. How does `exp2_throughput.awk` compute the throughput?
Ans. It sums the sizes of all received packets and divides by the time between the first enqueue and the last receive, multiplied by 8/1000 to obtain kb/s.

Q. Which link is the bottleneck here?
Ans. The link n2–n3, because the traffic of both flows has to pass through it.

Experiment 3 — Ethernet LAN: Throughput versus Data Rate and Error Rate

Q. What is CSMA/CD and how does a station react to a collision?
Ans. Carrier Sense Multiple Access with Collision Detection: a station listens before and during transmission; on a collision it stops, and retries after a random delay chosen by binary exponential backoff.

Q. What does the `make-lan` command create?
Ans. A shared LAN (one common channel) of the listed nodes, giving each node an LL, an interface queue and the specified MAC, here `Mac/802_3`.

Q. What is the purpose of `$tcp set class_ 1` and `$ns color 1 red`?
Ans. `class_` sets the flow id of the packets; `color` makes NAM draw the packets of flow 1 in red.

Q. Why does the throughput fall when the error rate is increased?
Ans. Lost packets must be retransmitted and TCP shrinks its congestion window after each loss, so less data is delivered in the same time.

Q. Why does the throughput hardly increase when the LAN rate is raised from 2 Mb to 3 Mb?
Ans. The LAN is no longer the bottleneck; the TCP window and the round-trip time limit the sending rate.

Experiment 4 — Congestion Window of TCP Vegas and Reno

Q. What is the congestion window?
Ans. The sender-side limit on the amount of unacknowledged data; it grows when acknowledgements arrive and is reduced when congestion is detected.

Q. Distinguish slow start from congestion avoidance.
Ans. In slow start the window doubles every round-trip time; in congestion avoidance it increases by about one packet per round-trip time.

Q. How do TCP Reno and TCP Vegas detect congestion?
Ans. Reno reacts to packet loss; Vegas watches the round-trip delay and adjusts the window before losses occur.

Q. What do `$tcp1 attach $tfile1` and `$tcp1 trace cwnd_` do?
Ans. They connect the agent to an open file and log the variable `cwnd_` to it every time it changes.

Q. What is a collision in Ethernet and how is it handled?
Ans. Two stations transmit at the same time and the frames overlap; CSMA/CD detects it, both stations stop, and each retransmits after a random backoff delay.

Q. Why is the Vegas curve flat after about 3 s?
Ans. Vegas has reached the window at which the queue it creates in the network stays between its thresholds, so it no longer changes the window.

Experiment 5 — Bit Stuffing in the Data-Link Layer

Q. Why is stuffing needed in the data-link layer?
Ans. To prevent the flag pattern that delimits a frame from appearing inside the data, so that the receiver finds the frame boundaries unambiguously.

Q. What is the HDLC flag and what is the bit-stuffing rule?
Ans. The flag is `01111110`; the sender inserts a 0 after every five consecutive 1s of data.

Q. How does the receiver recognise a stuffed bit?
Ans. A 0 that directly follows five consecutive 1s is a stuffed bit and is removed.

Q. What is the worst-case overhead of bit stuffing?
Ans. When the data is all 1s: one extra bit per five data bits, i.e. 20 %.

Q. How does byte stuffing differ?
Ans. It works on whole bytes: a flag or escape byte in the data is preceded by an escape byte (as in PPP).

Experiment 6 — Error Detection Using CRC (CRC-CCITT)

Q. What is a CRC and how is the check value calculated?
Ans. A polynomial code: the message with r appended zeros is divided by the generator polynomial using modulo-2 arithmetic and the r-bit remainder is the CRC.

Q. Write the generator polynomial of CRC-CCITT.
Ans. $x^{16} + x^{12} + x^5 + 1$ (`0x11021`, 17 bits).

Q. Why does a zero remainder at the receiver indicate no error?
Ans. The transmitted codeword is an exact multiple of $G(x)$; an error pattern $E(x)$ that is not a multiple of $G(x)$ leaves a non-zero remainder.

Q. Which errors can a CRC not detect?
Ans. Error patterns that are themselves divisible by the generator polynomial.

Q. How many zeros are appended before the division?
Ans. Degree of the generator, i.e. `len(div) - 1`: 3 here, 16 for CRC-CCITT.

Q. What is modulo-2 arithmetic?
Ans. Arithmetic without carries in which both addition and subtraction are the XOR operation.

APPENDIX B

Troubleshooting Guide

The messages below were produced by the real tools (NS-2 2.35, `gawk`, `mawk`, Python 3) for the mistakes that are easiest to make when typing the programs. If a program does not run in the laboratory, compare it line by line with this list.

Message or symptom	Cause	Fix
NS-2 / Tcl		
`can't read "tracefile": no such variable`	A variable used inside `proc finish` is not listed in `global`.	`global ns namfile tracefile` (Experiment 3).
`can't read "cwnd1": no such variable`	`$cwnd1.tr` is read as the variable `cwnd1` followed by `.tr`. The channels were opened as `tfile1`, `tfile2`.	`global ... tfile1 tfile2` and `close $tfile1`, `close $tfile2` (Experiment 4).
`invalid command name "MAC/802_3"`	Class names are case sensitive.	Write `Mac/802_3`.
`invalid command name "8"`	A stray character on a line of its own: Tcl treats the first word of every line as a command.	Delete the stray character.
No message, but the LAN behaves differently and the throughput changes	The `make-lan` command was broken over two lines, so the last arguments were not part of the command.	Keep the whole `make-lan` command on *one* line.
No message, but packets are 210 bytes instead of 500	`packetSize_` was written as two words (`packet Size_`); NS-2 does not complain.	Write `$cbr0 set packetSize_ 500`.
A `%` prompt appears and nothing runs	`ns` was started without a script.	Run `ns exp1.tcl`; leave the prompt with Ctrl-D.
AWK		
`BEGIN blocks must have an action part` (gawk), `syntax error at or near end of line` (mawk)	In AWK the `{` of a `BEGIN` block must be on the same line as `BEGIN`.	Write `BEGIN {`.
`syntax error` (gawk), `missing ) near ;` (mawk)	Unbalanced parentheses, for example in the `Tput` formula.	Make every `(` match a `)`.
`invalid char ''' in expression` (gawk), `unexpected character '''` (mawk)	AWK strings are written with double quotes only.	`printf("%f\t %.2f\n", time, Tput);`
The XGraph window or the data file is empty	The `printf` line of the AWK script is commented out with `#`, so nothing is printed.	Remove the `#` in front of `printf`.
Throughput is negative (about −6200 kb/s in Experiment 3)	The test is `time < stopTime` instead of `time > stopTime`, so `stopTime` never increases.	`if (time > stopTime)`.
Python		
`SyntaxError: expected '('` (older versions: `invalid syntax`)	A hyphen or a space in a name (`bit-stuffing`, `return destuffed data`).	Use underscores: `bit_stuffing`, `destuffed_data`.
`NameError: name 'FALSE' is not defined`	The Python constants are `True` and `False`.	`flag = False`.
`SyntaxError` at `if bit == ‘1’:`	A backtick was typed instead of a quote.	`if bit == '1':`

APPENDIX C

Files and Quick Command Reference

Files used in each experiment

Exp.	Programs and data typed by the student	Files produced
1	`exp1.tcl`, `exp1.awk`, `plot_graph_1`, `plot_graph_2`	`out.tr`, `out.nam`
2	`exp2.tcl`, `exp2.awk`, `exp2_throughput.awk`, `plot_tcp`, `plot_udp`	`out.tr`, `out.nam`, `plot_throughput`
3	`exp3.tcl`, `exp3.awk`	`exp3.tr`, `exp3.nam`, `exp3.graph`
4	`exp4.tcl`, `exp4.awk`	`exp4.tr`, `exp4.nam`, `cwnd1.tr`, `cwnd2.tr`, `tcp1`, `tcp2`
5	`exp5.py`	—
6	`exp6.py`	—

Commands

Open a terminal in the folder that contains the programs and type:

Terminal

```
# Experiment 1
ns exp1.tcl
awk -f exp1.awk out.tr
xgraph plot_graph_1
xgraph plot_graph_2

# Experiment 2
ns exp2.tcl
awk -f exp2.awk out.tr
xgraph plot_tcp
xgraph plot_udp
awk -f exp2_throughput.awk out.tr > plot_throughput
xgraph plot_throughput

# Experiment 3  (exp3.graph is written by the finish procedure)
ns exp3.tcl
xgraph exp3.graph

# Experiment 4  (tcp1 and tcp2 are written by the finish procedure)
ns exp4.tcl
xgraph tcp1
xgraph tcp2

# Experiments 5 and 6
python3 exp5.py
python3 exp6.py
```

Comparing several runs. Save the data file of every run under a different name (for example `cp exp3.graph rate_1Mb`) and give all of them to one XGraph command, `xgraph exp3.graph rate_1Mb`, to draw the curves in the same window.